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Turbulent Flames in Enclosed Combustion Chambers: Characteristics and Visualization—A Review

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Remarkable progress has been achieved in measuring the flame propagation rate accurately under laminar conditions, which can be used to predict turbulent flame propagation rates using some correlations fitted to experimental data. However, such propagation rates, unlike the laminar case, cannot be unambiguously determined. Nevertheless, the advancement of laser imaging techniques has led to several definitions of turbulent burning rates (Roshan et al., 2010, "Simulation of Global Warming Effect on Outdoor Thermal Comfort Conditions," Int. J. Environ. Sci. Technol., 7(3), pp. 571–580). Recently, a unified scaling factor has been successfully demonstrated using data gathered from several fan-stirred bombs. Such results are promising in compiling a comprehensive database of turbulent propagation rates for potential and common fuels of interest to internal combustion engines (ICEs) and gas turbines alike. The strict worldwide legislation to reduce emissions has forced many industries to look into alternative fuels with less emissions. One such alternative fuel that has gained much interest recently is the gas-to-liquid (GTL) fuel, which is being used in blended forms in several combustion applications. However, detailed combustion characteristic investigations are required before using this new alternative fuel widely in engines (Business, 2018, "Qatar's Exporters Directory 2018"). In this study, the significant issues associated with the use of fan-stirred bombs are investigated. First, the effect of varying fan speed and geometry is reviewed, and then, the measurement techniques that are commonly used to track flame propagation are discussed. This is followed by the study of the effect of using different types of fuels on combustion characteristics. Furthermore, the use of diesel and gasoline optical engine setups as advanced flame visualization tools have been reviewed extensively.
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1 Introduction

Remarkable progress has been achieved in measuring the flame propagation rate accurately under laminar conditions for several fuels and fuel blends including gas-to-liquid (GTL) fuel [1]. The laminar flame speed is one of the combustible mixture properties that depends entirely on the system temperature, pressure, and equivalence ratio Φ (both Φ and ϕ symbols are used in literature). The measured laminar flame speeds (corrected for flame-stretch and geometry-induced effects) can be used in the validation of chemical kinetic mechanisms. Besides, laminar flame speeds can be used to predict turbulent flame propagation rates using some correlations that are almost fitted to a particular experimental database. However, such propagation rates, unlike the laminar case, cannot be unambiguously determined. In spark-ignited engines, flame speed influences knocking events and plays an essential role in determining their performance and emission rates. Laminar burning velocity is the governing physicochemical parameter for the deflagration wave, indicating the diffusivity, reactivity, and exothermicity for a combustible mixture. Thus, the study of such parameter is fundamental to validate relevant kinetic mechanisms, which characterizes the combustion of future fuels. Furthermore, the study of both laminar and turbulent flame speeds is essential in modeling and designing efficient engines, in addition to other predictive computational modeling tools [1–5].

The advancement of laser imaging techniques has led to several definitions of turbulent burning rates [6]. Recently, a unified scaling factor has been successfully demonstrated using data gathered from several fan-stirred bombs. In addition, a self-similar scaling for spherically expanding flames could be effectively derived [7,8]. Such results are promising in compiling a comprehensive database of turbulent propagation rates for potential and common fuels of interest to internal combustion engines (ICEs) and gas turbines alike. Fan-stirred bombs are commonly used to study premixed turbulent flame geometry. These vessels allow the flame propagation rates to be measured conveniently under turbulent conditions, where a wide range of intensity levels u' can be obtained, and they are well controlled. This feature is exceptionally advantageous in comparison with a burner-type setup, as the latter requires an increase in the level of turbulence intensity by an additional increase in the axial flowrate. When the flowrate is high, the use of stabilizing mechanisms such as recirculation zones, pilot flame, or the bluff body becomes necessary to prevent flashback. Nevertheless, the design of fan-stirred vessels is cost-effective and does not allow laser-based planar measurements to be used flexibly [6].

In the past 20 years, there has been a great deal of research and effort on minimizing environmental damages due to global warming. Human activities like the excessive use of fossil fuels can promote global warming and accelerate climate change destructively, and there is a need to control it [9]. The strict legislations in

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the European Union and in the USA to reduce emissions have forced many industries to look into alternative fuels with less emission production. One such alternative fuel that has gained much interest recently is the GTL fuel.

Currently, Qatar is the largest producer and exporter of GTL products, with a production rate of 140,000 barrels per day of high-quality GTL products [10]. The GTL fuels have not replaced conventional fuels yet entirely, but they are in use in blended forms in several combustion applications, mainly ICEs. This puts a responsibility on researchers to find out the optimum conditions and characteristics of GTL fuel, and to examine its suitability for use in ICEs. GTL fuel can provide a solution for the high emissions level of combustion engines. However, detailed combustion characteristic investigations are required only before using this new alternative fuel widely in engines.

In this literature review study, the significant issues associated with the use of fan-stirred bombs for turbulent flame speed measurements and characterization are reviewed and discussed. First, the effect of varying fan speed and geometry is looked at in detail. After that, the measurement techniques that are commonly used to track the flame propagation are discussed. This is followed by the study of the effect of using different types of fuels on combustion characteristics. Furthermore, this literature review aims to investigate in detail the use of diesel/gasoline optical engine setup as an advanced flame visualization tool.

2 Combustion Bomb

The use of fan-stirred bombs is one of the measurement techniques for investigating premixed turbulent flame geometry. These vessels allow the flame propagation rates to be measured conveniently under turbulent conditions, where a wide range of intensity levels u' can be obtained, and they are well controlled. In this literature review, the significant issues associated with the use of fan-stirred bombs are discussed as follows. First, the effect of varying fan speed and geometry is reviewed and discussed. After that, the measurement techniques that are commonly used to track the flame propagation are discussed. This is followed by the study of the effect of using different types of fuels on combustion characteristics.

The first use of fan-stirred vessels for the measurement of turbulent flame speed S_T (in literature, both S_T and S_t symbols are used for turbulent flame speed) is credited to Semenov [11]. Later, many researchers have considered the fan-stirred vessels to conduct their experiments. In the spherical combustion vessel, the flame initiates at the center of the vessel, and it propagates radially outward in all directions with uniform turbulence. Laser tomography, high-speed imaging cameras, and pressure trace measurements are three of the most commonly employed methods to diagnose the flame propagation rate [12]. With the use of this configuration, several advantages can be offered in comparison with the burner type, where the intensity level has to be achieved with a high mean velocity, in addition to some difficulties in stabilizing high laminar flame speeds S_L at the top of the burner surface [13]. The use of fan-stirred vessels overcomes all of these disadvantages, which are correlated with the system as it allows for the measurement of flame speeds in near homogeneous and isotropic turbulence (HIT) conditions with the use of high values of u'. On the other hand, the spherical combustion chamber suffers from some deficiencies as it is extremely challenging to develop them, and it is prohibitive to build them besides being very expensive. Furthermore, the problem of time-varying flame propagation is accompanied by the absence of a well-defined surface, which can be treated as a reference for the measured burning velocity [14]. Nevertheless, some previous studies claimed that these difficulties could be counteracted by making some appropriate assumptions [15,16]. For premixed turbulent combustion, the two major parameters used to characterize such flow fields are the integral length scale L_T and the turbulence intensity u'. The integral length scale is defined as the largest eddy that exists within the flow, while the turbulence intensity is used to characterize the level of turbulence. In this section, the effect of varying the fan speed and geometry is discussed first. After that, the discussion is dedicated to the effect of using different turbulent flame speed measurement techniques to track the flame propagation. Finally, the effect of varying the fuel type on the combustion characteristics is handled in the last subsection.

2.1 Fan Speed and Geometry. This subsection aims to review and study the literature on the effect of varying the fan speed on maintaining HIT condition and turbulent flame speed S_T. In addition, the effect of varying the fan geometry such as the pitch angle and the number of blades is studied as well.

2.1.1 Fan Speed. There is a clear contribution to the variation of turbulence intensity u' on the turbulence flame speed S_T. In the beginning, S_T starts to increase with u' until it reaches a maximum level; after that, it drops down until the flame quenching occurs. When the turbulence intensity level u' is lower than the laminar flame speed S_L ($u' < S_L$), the flame wrinkling that results from turbulence is dominated by flame propagation, and this regime is not considered relevant to any industrial system. On the other hand, at higher turbulent intensity levels ($u' > S_L$), there is a significant increment in the flame speed due to the enhanced mass and heat transfer rates that result in the turbulent diffusion. Most of the apparatus, which were used to study the turbulent flame speed in the combustion vessels, emphasize that there is a uniform distribution of velocity fields in the center of the vessel. However, two studies have revoked these two observations [17,18] and indicated that there is a non-linear distribution of velocity fields exists in the spherical combustion bomb. The observation of these anomalies can be referred to some specificities of the measurement setup; nonetheless, it is highly important to maintain HIT conditions within the spherical combustion vessel having a radius that is equal to the maximum size of the flame.

Birouk et al. [19] have studied the effect of fan speed variation on turbulence intensity and the integral length scale under HIT conditions. They have observed a gradual increase in u' when increasing the fan speed and turbulent kinetic energy (TKE) that ranges between 0.1 and 1.45 m^2/s^2. The instantaneous velocity probability density function (PDF) had a zero skewness (a measure of the symmetry of the probability distribution for a random variable about its mean), and a kurtosis (a measure of whether the data are light-tailed or heavy-tailed with respect to a normal distribution) of value three at all fan speeds, which further confirm that HIT conditions have been successfully achieved inside the chamber. In addition, the isotropy and homogeneity ratios have been computed and found to be within ±5% of the ideal value required to maintain a perfect HIT condition. The integral length scale could be obtained by integrating the lateral spatial correlation coefficient curve (was found to be 8.6 mm) and, unlike the turbulence intensity, L_T was observed to be constant at all fan speeds. Hwangand Eaton [20] have replaced the fans by eight pulsed-jet actuation system in a similar Plexiglas chamber and under HIT conditions. These fans were controlled at a precise rate to generate the required isotropy and homogeneity. A spatial map inside an area of 1600 mm^2 in the middle of the chamber was generated. It was found that the isotropy ratio varies between 0.88 and 1.24, and the homogeneity ratio had a fluctuation of ±10% of their spatially averaged value. The turbulent kinetic energy dissipation rate has been estimated using a large eddy simulation (LES) technique, and thus, other turbulence parameters such as Kolmogorov, Euler, and Taylor scales could be evaluated [21]. A considerable variation in the previous turbulence statistics was observed due to errors in evaluating the velocity derivatives. In addition, it was observed that the integral length scale, however, remains the same and spatially distributed in the space at all fan speeds.

2.1.2 Fan Geometry. Kwon et al. [22] varied the impeller's pitch angle from 45 deg (primarily used by Fansler and Groff

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[23]) to 30 deg and observed a decrement of 50% in L_T noting that the turbulence intensity level u' was maintained the same in the two cases. It was concluded that the impeller geometry affects the size of the turbulent eddies, therefore influencing the scale of L_T; however, HIT conditions have not been appropriately met due to misalignment in the impellers. In another later study [15], the misalignment of the impeller inside the combustion vessel has been considered and fixed; therefore, the mean flow has reduced from about 0.3–0.6u' to around 0.1u', which is almost a negligible mean flow. Leisenheimer and Leukel [17] used two vessels, which have different sizes (ϕ 50 cm and ϕ 130 cm, where ϕ is the diameter of the vessel) to vary L_T, and concluded that L_T depends mainly on the dimensions of the vessel rather than on the geometry of the impeller. Shy et al. [24] have considered the use of a perforated plate located in front of the impellers to eliminate any large vorticity formation in the flow and thus promote a rapid mixing. Liu [25] has studied the effect of such plates on the downstream turbulent field flow that takes place inside a wind tunnel. He has concluded that both L_T and u' can be varied individually by changing the orifice size in addition to the solidity ratio (a ratio between the area fractions of the plate's solid portions). In another study [26], four high-speed central-symmetric fans were installed inside a spherical combustion bomb, and three different fan designs that have radial shape instead of an axial one have been used in order to find the optimum impeller design to generate isotropic and homogeneous turbulence inside the combustion bomb. It was noticed that there is no remarkable variation in the intensity level when using different impeller designs, and the velocity probability density function for the three models was found to follow a Gaussian profile. The homogeneity of the flow field was found to be slightly dependent on the geometry of the impeller. However, the isotropy was observed to be sensitive to the geometry of the fan. As the number of blades increases (six blades instead of four), there is a higher deviation observed from the isotropic turbulence. As was discussed by Kwon et al. [22], the variation in blade pitch angle affects the integral length scale. However, this variation in L_T can occur while maintaining the near HIT conditions within the combustion vessel.

2.2 Measurement Techniques. In this subsection, the various techniques employed for turbulent flame speed measurements are discussed.

2.2.1 Schlieren Imaging. Schlieren imaging is one of the most common techniques used for laminar flame speed measurement and can be used to conduct turbulent combustion studies. In the laminar case, the flame starts to propagate outwardly after ignition in the radial direction. Consequently, the flame can be assumed to propagate in one direction and the Schlieren techniques despite seen as an integrated line-of-sight measurement method; it can be applied for configuration of such geometry. In the turbulent case, flame wrinkling results in a high anisotropy even in a well-controlled HIT environment, such as in the spherical combustion chamber. Thus, the propagation speed in the axial (out-of-plane) and the radial (in-plane) directions is not the same for any planar kernel. Nevertheless, it was demonstrated by Bradley et al. [14] that based on the kernel area found inside the Schlieren silhouette, the average radius corresponds to a reaction progress variable $\langle c \rangle \approx 0.1$ ($\langle c \rangle$ is an isotherm that indicates for the percentage of combustion completion determined at the surface). Different configurations of Schlieren setup can be used depending on various factors such as the available space, the experimental setup, and the objective behind the imaging technique. However, most of the past researches have considered the use of *Z*-type Schlieren setup [26–28]. This setup consists typically of two aluminized mirrors of spherical shape, a high-speed camera, a sharp blade knife-edge, and a single light-emitting diode (LED) spotlight. Typically, in this configuration, the LED light beam is emitted from the light source and is directed into the first mirror that is located in the same line of the LED source. After that, the reflected beam passes through the spherical chamber and continues toward the second spherical mirror where the light converges at the second mirror focal point until it hits the knife-edge where the dark background light white is cut off, and the rest of the light continues traveling to the high-speed camera. Figure 1 shows a schematic diagram for the Schlieren setup.

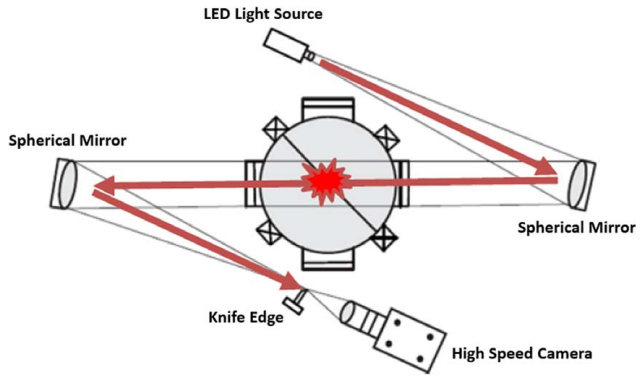

Fig. 1 Schematic diagram for *Z*-type Schlieren setup [28]

2.2.2 Pressure Trace. The second most widely used diagnostic technique for the measurement of flame speed is the use of dynamic pressure trace of high frequency [28]. This technique is inexpensive and does not need any optical ports to access the flame propagation. Inside the constant-volume vessel, the pressure keeps constant for a certain period, and then isentropic compression occurs for the unburnt gases before the flame initiates, which causes a pressure rise. However, the significant pressure rise is not reached until the flame propagates to a large diameter. Although this setup has its advantages, some deficiencies are associated with its use [14]. First, it is challenging to keep the spherical shape at large flame radii due to the confinement effects of the chamber. Second, the uniformity of turbulent conditions can no longer be maintained at large radii (close to the fans). Third, the isentropic compression of the unburned gas results in a large increment in the temperature and pressure; thus, flame speeds are not captured at the initial unburned premixed conditions. Moreover, flame visualization is not available when using this technique, which makes it an unattractive measurement technique. This technique was employed by Samim [1] for the measurement of GTL laminar flame speeds in a cylindrical combustion bomb, which has been designed at Qatar University. Although the vessel was designed for high-speed imaging through the quartz glass, only a normal video camera that captures 60 frames per second was used. The pressure data collected from the pressure sensor were used primarily to measure the flame speed using some equations designed for that purpose, which were proposed by Lewis and Von Elbe [29]. The proposed model has also been used in another similar research to compare the flame speed of Jojoba methyl ester with diesel fuel using a cylindrical vessel and relying on pressure transducer signals [30]. In addition, this technique has been employed by Ruelas et al. [31] to measure the laminar flame speeds for natural gas/hydrogen blends.

2.2.3 Laser Tomography. The instantaneous flame surfaces can also be determined using 2D planar, single-sheet laser imaging for the measurement of burning velocity. Nevertheless, there are many drawbacks associated with their use [26,32]. It is still unclear whether the assumption of considering the in-plane and out-of-plane rates to be similar for turbulent flames, and this issue is still questionable. In addition, the spark which initiates in the middle of the vessel is randomly convicted by large eddies, which results in a kernel displacement away from the laser-sheet imaging path. Consequently, the exact location of the kernel relative to the laser sheet cannot be accurately determined. Schlieren photography technique is more preferred than laser-sheet techniques

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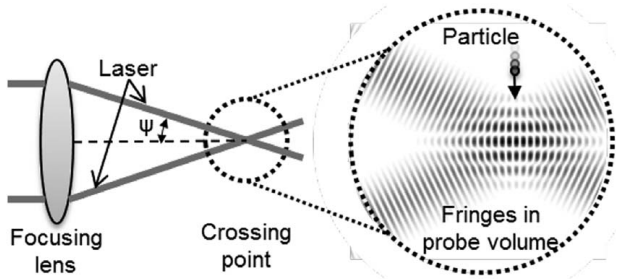

Fig. 2 Schematic diagram for the LDV setup [39]

because they almost give the same accuracy for flame propagation rates besides being less expensive and requiring less equipment [33–35].

One of the standard techniques used in laser tomography for measuring the velocity of particles inside the combustion vessel is the laser doppler velocimetry (LDV). In this technique, two laser beams are directed and focused on one point according to the desired velocity magnitude. The two laser beams share the same wavelength, and their intersection creates a volume of interference called the probe volume [28]. This technique was employed by Refs. [36–39] to confirm the particle image velocimetry (PIV) measurements with those of LDV in a spherical combustion vessel, where the interrogation area in the LDV system is expanded here into a 3D volume. Figure 2 shows a schematic diagram for the LDV setup.

2.3 Fuel Type. As the fuel characteristics (reactivity, diffusivity, and exothermicity) play a significant role in determining the fuel-air combustion characteristics [40–42], it is with high importance to study the effect of using different types of fuels on the flame propagation rates. Several experiments were conducted in literature utilizing the spherical/cylindrical combustion bomb to match this demand. In this context, the role of this subsection comes to look at the most recent researches that have investigated the effect of different fuel properties on the mixture combustion characteristics; namely, four types of fuels are included in this literature: alkanes, natural gas blends, isooctane, and biogas.

2.3.1 Alkanes (C_1-C_3). Turbulent flame speeds for alkanes, particularly for methane, ethane, and propane, were measured over a wide range of equivalence ratios by Ravi [26]. It was observed that turbulent flame propagation is profoundly affected by non-unity Lewis numbers, Le (Le is a dimensionless number that represents the ratio between thermal diffusivity and species mass diffusivity); such that displacement speeds were noticed to be higher for mixtures with $Le < 1$ in comparison with those of $Le > 1$ at the same equivalence ratio. Subsequently, methane was characterized by a higher turbulent flame speed than ethane and propane when running at lean conditions. However, rich methane had a lower displacement speed than ethane and propane. Thus, it was concluded that for mixtures characterized with $Le < 1$, the displacement speeds are augmented by an increment in both flamelet surface area and local burning rate. Figure 3 shows the variation in turbulent flame speed as a function of turbulence intensity for methane and ethane.

2.3.2 Natural Gas and Natural Gas–Hydrogen Blends. As the natural gas composition varies depending on the origin of the fuel source, turbulent displacement speed for a natural gas surrogate (NG_2: 1.25% n-C_5H_{12} + 2.5% C_4H_{10} + 5% C_3H_8 + 10% C_2H_6 + 81.25% CH_4) has been investigated by Ravi [26] over a wide range of equivalence ratios. The displacement speeds for the C_1–C_3 alkanes and NG_2 were all compared by using the same combustion vessel under different equivalence ratios Φ (Fig. 4). When $\Phi = 0.7$, the turbulent flame speeds for methane were the highest, followed by ethane and then propane. At this condition, NG_2 is noticed to have a very close trend to CH_4, as NG_2 is predominantly methane. On the other hand, rich methane is characterized by lower displacement speeds than rich ethane and propane, and this can be referred to the impact of Lewis number on turbulent flame propagation. When $Le > 1$ ($Le < 1$), the local burning velocity is reduced (enhanced) due to the enhanced (reduced) heat loss in the sections which are positively curved [43]. The probability of finding curved flamelets, positively stretched (convex into the direction of unburnt gases), is higher with a premixed turbulent reacting flow as the mean curvature and the mean strain rate are both positive at the flame leading edge [44]. In this case, turbulent eddies cause more stretching in the laminar flamelets, which results in a remarkable deviation between the unstretched laminar flame speed and the local burning rate. Subsequently, for mixtures having $Le < 1$, the displacement speeds are augmented by an increased local burning rate and increased flamelet surface area. These effects interpret the higher turbulent flame speeds for rich ethane and rich propane (or lean methane), which have $Le < 1$. Such trends have also been previously documented by other studies [18,44].

Blending hydrocarbon fuels with hydrogen is essential for gas-turbine combustor designers. This is due to the higher laminar flame speeds, which can be obtained at ultra-lean conditions, which are outside hydrocarbons lean-flammability limit. However, the tendency to flashback or auto-ignite increases as the fraction of hydrocarbons in the mixture increases as a result of the augmented laminar flame speeds (the turbulent flame speeds as well).

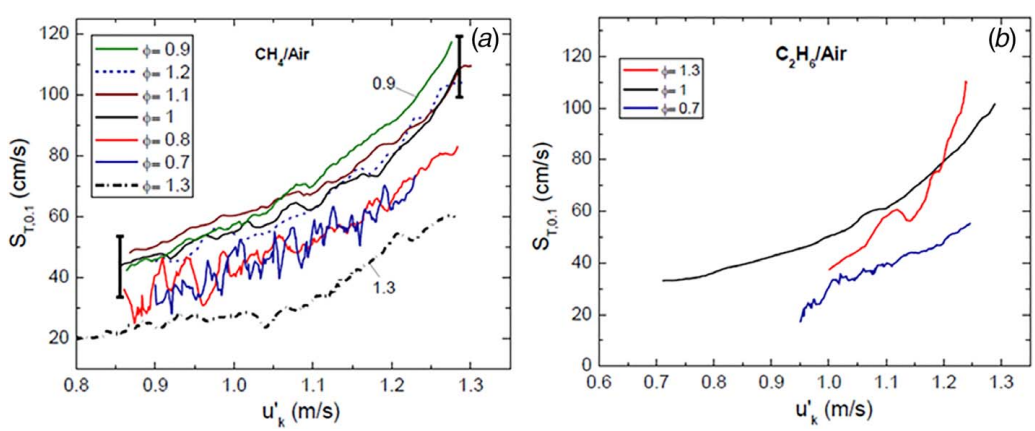

Fig. 3 Turbulent flame speeds of C_1, C_2 alkanes/air mixtures as a function of the effective turbulence intensity. The individual curves are the spline averages of three repetitions at each condition. The average scatter is estimated to be ±10 cm/s [26].

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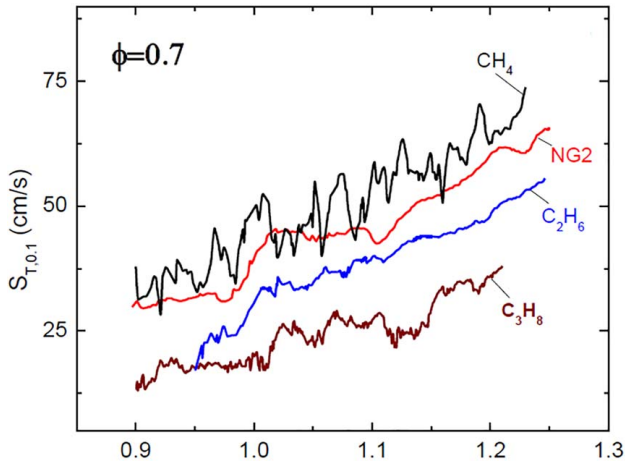

Fig. 4 Displacement speeds of alkanes and NG_2 at the fuel-lean condition. Mixtures with Le >1 have lower turbulent flame speeds (lean-ethane and propane, and rich-methane) due to the reduction in the stretched laminar flame speeds from the unstretched values [26].

The increased concentration of hydrogen radicals in the mix is the primary mechanism for laminar flame speed augmentation, which is purely a kinematic effect. Laminar and turbulent flame speeds for two different mixtures (25/75 and 50/50 (by volume) blends of H_2/NG_2) at different equivalence ratio Φ and hydrogen levels was investigated by Ravi [26] and are plotted in Fig. 5. It can be observed that there is no difference in the laminar flame speeds between CH_4 and NG_2 for the case of high-hydrogen content (50% H_2). In addition, it is clear how the turbulent flame speed S_T has increased for both fuels with hydrogen addition, and it is following the same trend found with the laminar flame speeds.

2.3.3 Isooctane. Brequigny et al. [36] have experimentally investigated turbulent flame speeds for isooctane–air mixture. Figure 6 [36] illustrates the corrected turbulent flame speed as a function of the corrected equivalent radius. As can be noticed, the flame speed is increasing gradually while the flame radius is increasing, and this can be referred to the increase in the wrinkling ratio. The increase in turbulence intensity is responsible for the rise in wrinkling and, thus, the turbulent flame speed. Typically, the increase in wrinkling ratio is provoked by a decrease in small, turbulent scales, as was already shown by Mazellier et al. [37] on the same combustion vessel.

2.3.4 Biogas. Turbulent flame speeds for three different biogas surrogates have been experimentally investigated by Ayache and Birouk [28]. It was observed that when adding more CH_4 into the mixture, the peak value for S_T (= 50 cm/s) shifts away from the nearly lean equivalence ratio side into the stoichiometric value (Fig. 7). In addition, it can be noticed that all the curves are "bell-shaped" where the maximum burning velocity occurs at a certain equivalence ratio and diminishes at the higher and lower ones [28]. The slowest burning velocity among all the surrogates mixtures was observed when using the 50% CH_4 biogas flame, with a peak burning velocity of 27.5 cm/s at $\varnothing = 0.9$. When more CH_4 is added into the mixture, the peak has shifted away slightly toward the stoichiometric equivalence ratio, where the peak was noticed at 39 cm/s at $\varnothing = 0.98$ for 70% CH_4. For the mixture (70% CH_4 and 30% CO_2), the rich and lean turbulent flame speeds were faster than those of 60% and 50% by 5 cm/s.

2.3.5 Syngas. Different researches have investigated the laminar flame speeds for syngas over the past decades [45–51]. Among these large databases, the laminar flame speeds for syngas have been reported at the same operating conditions and then compared together, as shown in Fig. 8. It is clearly noticed that large disparities exist between the measured laminar flame speeds for syngas/air mixtures at normal temperature and pressure, and those calculated by updated chemical reaction mechanism by Li et al. [51]. Also, it can be noticed that the curve of laminar flame speeds for syngas is "bell-shaped" where the maximum burning velocity occurs at a certain equivalence ratio ($\Phi = 2.15$) and diminishes at the higher and lower ones. Li mechanism slightly overpredicts the whole measurements at lean operating conditions. However, when $\Phi > 2.0$, large disparities exist between the measurements of different researches. When ϕ increases more, the discrepancy becomes even larger. This is considered as an important issue as the accuracy of laminar flame speeds measurement plays a major role in building and validating chemical kinetic models.

The reason behind this discrepancy among these different groups can be attributed to several reasons as follows. The first one is the difference in spherical flame radius ranges which is chosen for nonlinear or linear extrapolation. Subsequently, this instills a responsibility on the researches who should determine the lower and upper bound accurately beforehand. The second reason is the methodology that is used to extrapolate laminar flame speeds readings. For example, the stretch rate can highly influence the flame speed when the laminar flame propagates in the spherical combustion chamber and thus causes a remarkable difference in the obtained results. Another possible reason is the existence of carbonyl compounds inside CO_2 gas cylinder [52].

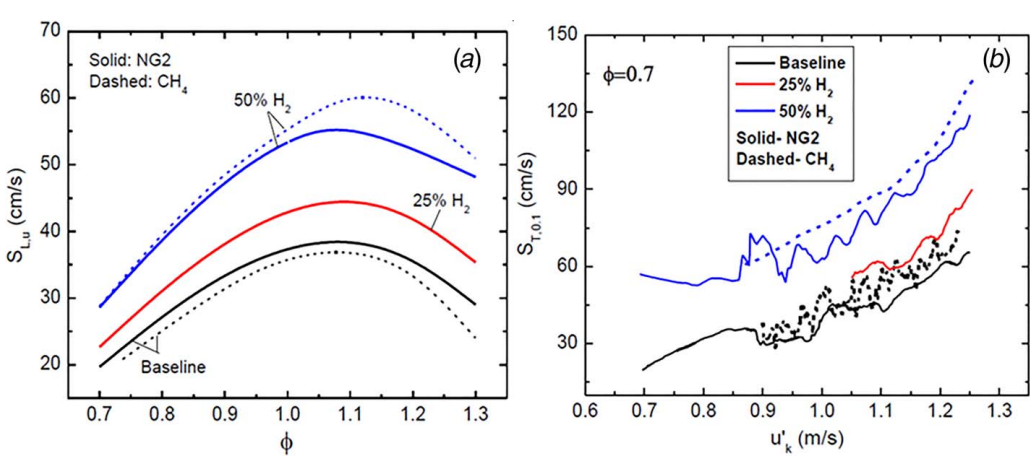

Fig. 5 (*a*) Laminar flame speeds (curves are model predictions) of NG_2 and CH_4 with varying levels of H_2. (*b*) Turbulent displacement speed for these mixtures measured herein. Turbulence intensity was estimated separately for each fuel. No appreciable difference is observed between the high-hydrogen-content mixtures of methane and NG_2 [26].

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2.3.6 Diesel. Laminar flame speeds for diesel fuel have been investigated by Samim [1] using a cylindrical bomb test rig that has been designed at Qatar University. It was observed that the flame speed is slightly low at lean conditions ($\Phi = 0.7$–0.9), which further increases and peaks near stoichiometric conditions ($\Phi = 1.0$–1.1). When the mixture becomes rich ($\Phi = 1.2$–1.3), the flame speed starts to decrease slightly below the maximum laminar flame speed ($S_L = 83$ cm/s) as indicated in Fig. 9. At low equivalence ratios, the excessive amount of air causes flame quenching, and the flame becomes unstable. As a result, the laminar flame speed is found to be low. On the other hand, using an excessive amount of fuel in the air–fuel mixture causes the combustion to be incomplete which causes the flames speed to drop down.

2.3.7 Gas-to-Liquid Fuel Blends. Laminar flame speeds of GTL, diesel, and 50/50 diesel-GTL fuel blends have been measured over a wide range of equivalence ratios by Samim [1], as shown in Fig. 10. It was observed that GTL fuel has the highest laminar flame speeds near stoichiometric conditions ($\phi = 1.0$), which might be attributed to its highest calorific value. However, when the mixture equivalence ratio becomes further away from the stoichiometric condition, the flame speed of GTL gets slightly lower than pure diesel. Thus, it can be deduced that GTL fuel can replace diesel in applications where the operating conditions are close to stoichiometric condition, and as such, this will result in a higher fuel laminar flame speed.

The 50/50 blend starts with the lowest laminar flame speed at the low equivalence ratios ($\Phi = 0.7$–0.9) and maintains this behavior until it equalizes the readings of conventional diesel at $\Phi = 1.0$. When $\phi = 1.1$, the flame speed for the 50/50 blend is only higher than the conventional diesel by 2–3 cm/s. After that, the flame speed starts to drop lower than both fuels. It can be observed that the 50/50 blend is exhibiting lower flame speeds than both pure

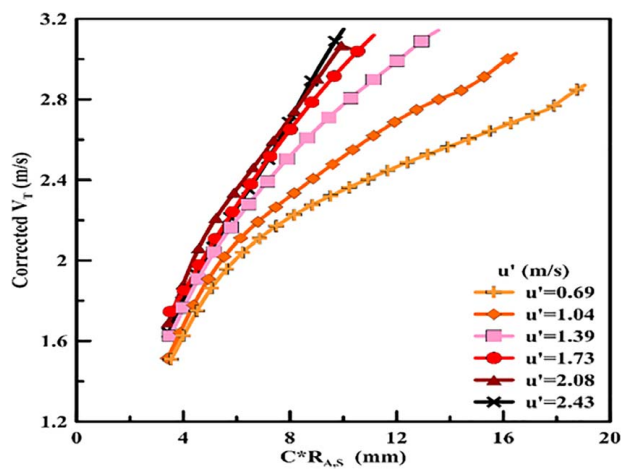

Fig. 6 Corrected flame speed as a function of flame radius for different turbulent intensities obtained from the corrected Schlieren data: $P = 0.1$ MPa, $T = 423$ K, and $\Phi = 1.0$ [36].

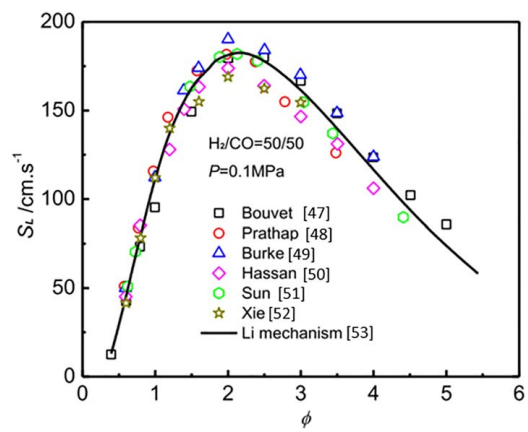

Fig. 8 Laminar flame speeds of H_2/CO mixtures at atmospheric pressure and standard temperature: comparison between experimental data sets and numerical predictions [45–51]

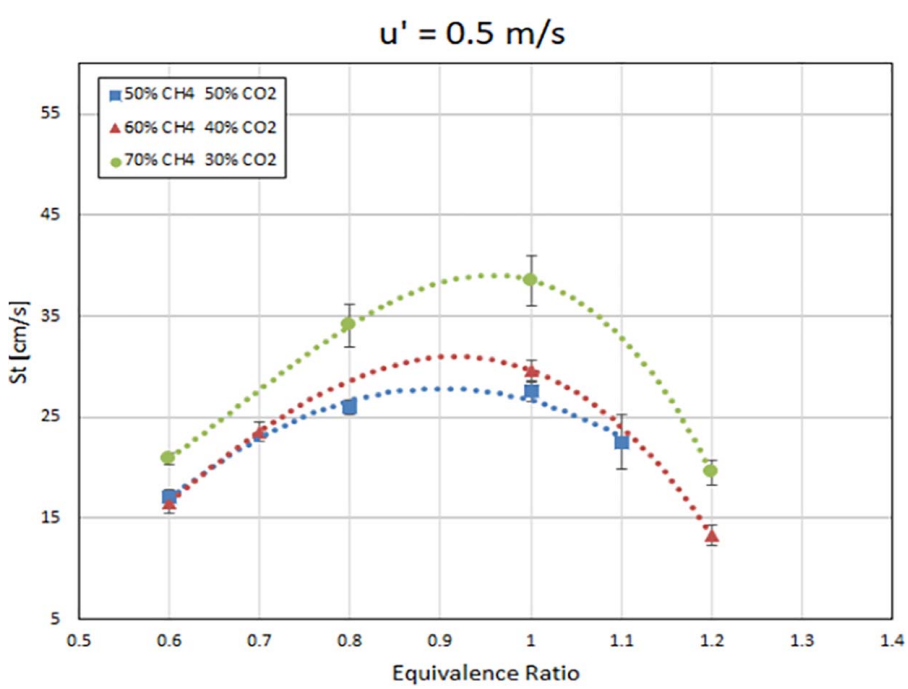

Fig. 7 Biogas turbulent burning velocity for three biogas compositions at fixed turbulence intensity. For all compositions, the curve fits of the experimental data are "bell-shaped" with the burning velocity peaking at some equivalence ratios and diminishing at lower and higher ones [28].

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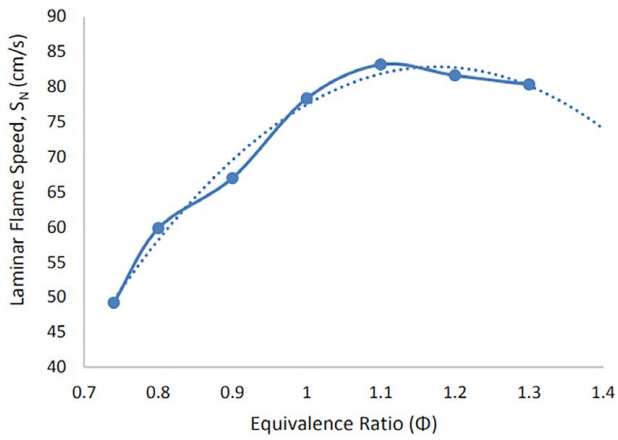

Fig. 9 Laminar flame speeds of diesel fuel at different equivalence ratios [1]

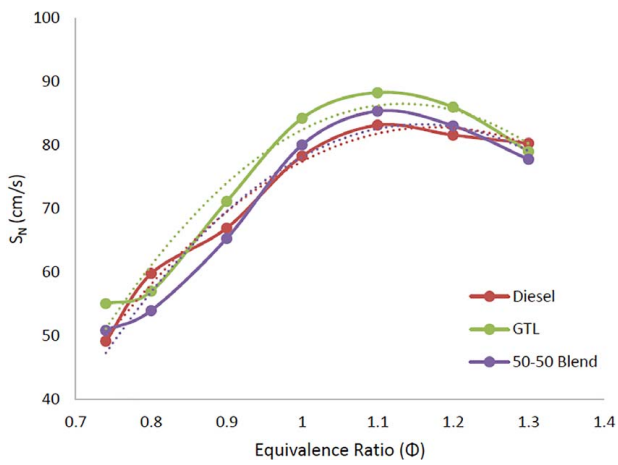

Fig. 10 Conventional diesel, GTL, and 50–50 blend (laminar flame speed, S_N versus Φ) [1]

fuels away from the stoichiometric conditions. However, near stoichiometric conditions, the readings for the 50/50 blend are very close to those of diesel fuel, which indicates that this blend can replace diesel fuel in industry to reduce exhaust emissions and keeps the environment cleaner. For the three tested fuels, the readings for the laminar flame speed at the lean conditions (e.g., excess air) is observed to be low due to flame quenching. On the other hand, the excessive use of fuel at rich conditions causes the combustion to be incomplete which results in a lower flame speed.

Discussion

Laminar flame speed S_L is defined as the propagation velocity of the flame front into the premixed unburned gas mixture. It was clearly noticed that using different types of fuels results in a remarkable variation in the flame speed (e.g., GTL has a higher laminar flame speed than diesel fuel at different equivalence ratios). In addition, it was observed that the laminar flame speed reaches its maximum value near stoichiometric conditions for hydrocarbon fuels such as diesel, GTL, and 50/50 diesel-GTL fuel blend. Furthermore, the laminar flame speed depends on other factors such as the temperature and pressure inside the combustion vessel.

Turbulence causes flame propagation to be promoted, causing the flame speed to be considerably higher than the laminar one. The impact of turbulence intensity on the flame speed is also dependent on the stoichiometry and on the fuel type. The effect of using different types of fuels is often explained by the impact of molecular and thermal diffusivity for the fuel on the evolution of corrugated flamelets, which can be captured by Lewis number (a ratio between thermal diffusivity and molecular diffusivity). For example, it has been noticed how methane and natural gas are characterized by a higher turbulent flame speed than ethane and propane when running at lean conditions. However, rich methane had a lower displacement speed than ethane and propane (Fig. 4). Also, it was observed how the mixtures of higher hydrogen content are characterized by a higher turbulent flame speed (Fig. 5). This can be referred to the lower Le number of hydrogen when compared with other hydrocarbons. Moreover, it was noticed how the increase in turbulence intensity causes the flame speed to be augmented due to the increase in the wrinkling ratio, which results in more turbulence within the mixture (Fig. 6).

In an internal combustion engine, the flame speed is one of the fuel properties which determines its ability to be combusted without detonation. Subsequently, a high-flame speed combustion process should reflect in higher engine efficiencies and lower emissions due to the enhanced flame diffusivity and stability, which can thus result in more complete combustion. In fact, the speed at which laminar or turbulent premixed flames propagate is a fundamental parameter that is used in many practical combustion applications, such as gas turbines and engines to validate relevant kinetic mechanisms, which characterizes the combustion behavior of the fuel [30,31].

An accurate description of the combustion phenomena is highly requested to assess the effect of using a certain type of fuel on the engine performance. Besides, this can highly assist in the development of advanced engine concepts for alternative fuels and petroleum-derived. Diesel, gasoline, and jet fuels are composed of several chemical compounds, and thus, making detailed predication and modeling for real fuels is very complex. In addition, the lack of fundamental thermochemical, physical, and kinetic data for the components found in real fuels makes the study of flame speed very essential to predict engine performance and emission rates [28].

Summary

Table 1 provides a summary of literature studies of vessel shape and geometry, measurement techniques, fan details, turbulent characterization techniques, and fuel type.

3 Optical Engine

The use of thermodynamics single-cylinder engines (TSCE) in Research and Development (R&D) applications provides a versatile environment for testing the primary process and modules related to engine operation. Diagnostic instrumentation of optical single-cylinder engines (OSCE) is less complicated as compared to a multi-cylinder engine (MCE), and the simpler mechanical structure provides more flexibility in applying refined diagnostic access [53]. OSCEs offer specific opportunities for higher diagnostics of engine phenomena. This has been considered as a driving motivation to engage them in research projects. As modern engine systems are being refined, and their complexity is growing when achieving the required emission targets and fuel efficiency, it becomes vital to use OSCE in the application of research projects. Moreover, diagnostic instrumentation of OSCE is less sophisticated in comparison with other MCEs, and the simple mechanical design offers more flexibility when applying refined diagnostic access [54]. Since OSCE can only duplicate the behavior of MCE partially, care should be taken when deciding the most influential usage of OSCE in engine development tasks. In this aspect, the priority for using OSCE comes in providing data, which can depict the behavior of a normal MCE such as in fuel injection and the simulation of intake airflow. Both applications of research and development benefit from each other as the use of optical engines promotes progress, and new analysis techniques are occupied. OSCE is

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Table 1 Summary of literature survey on spherical and cylindrical combustion bombs

Reference	Fansler Groff [23]	Kwon et al. [22]	Leisenhei-mer and Leukel [17]	Weib et al. [18]	Shy et al. [24]	Bradley et al. [14]	Ravi [26]
Vessel shape	Cylindrical	Cylindrical	Spherical	Cuboid with spherical cavity at center	Intersection of two cylinders	Spherical	Cylindrical
ID (cm)	26	26	Vessel 1 (V1): $d = 50$	11.8	Cubic volume of $15 \times 15 \times 15$ cm^3	38	30.5
IL (cm)	26	26	Vessel 2 (V2): $d = 130.2$	11.8		38	35.6
No. of optical ports	2	2 + 2 Laser ports	2	4	4	3	2
Optical port Size (cm)	9.2 (8 cm: max. flame dia.)	9.2; 10 mm	Not provided	10	10	15	12.7
Meas. technique	Schlieren imaging	Schlieren imaging and laser tomography	Pressure trace	Pressure trace	Laser tomography	Schlieren imaging and pressure trace	Schlieren imaging and pressure trace
No. of fans	4	4	V1: 2,4,8; V2: 4	8	2	4	4
Fan dia. (cm)	13.5	13.5	V1: 25; V2:42	4.5	11.6	Not provided	7.62
Fan blades	8		Not provided	6	8	8	3
Pitch angle (deg)	45	30	Not provided	22.5	Not provided	Not provided	20
Fan width (cm)	2.3		Not provided	0.6	4	Not provided	3.8
Turb. Char. Tech.	LDV	LDV	Five-hole pitot probe and Hot wire Anemometer	LDV/PIV	LDV/PIV	LDV	PIV
Max u' (m/s)	2		V1: 1.54 m/s; V2: 2.4 m/s	~3.7	4.5	12	3.5
Longitudinal L_T (mm)	25 mm w/u'; 40 mm using 3D Gaussian u'	12.5; 19.15 mm with 3D Gaussian u'	V1: 9 ± 1 mm; V2: 24 ± 2 mm	3.9	15–48	20	50
Fuel	Not provided	Hydrogen + air + nitrogen	Not provided	Methane–air and propane–air and hydrogen	Methane–air	Ethanol–air and propane–air	Alkanes (C_1-C_3) and natural gas and natural gas–hydrogen blends

combined with large optical windows that allow for optical imaging and a large field of view through the combustion chamber [55]. The OSCE consists of different elementary design components that are accompanied by a photography system for the gasoline direct injection (GDI) engine and elongated piston modules [56]. The extended crankcase design is employed to access the combustion chamber from the bottom side using a 45-deg mirror. Besides, it allows for the separation of the base engine's lube oil from the "dry" zone of the combustion chamber [57]. Figure 11 shows a schematic diagram for the major components that a single-cylinder optical engine consists of.

In general, optical engines are known to have high flexibility in their operation due to the separation of many peripheral modules like the coolant pumps, heat exchangers, and fuel supply systems in addition to the lube oil from the engine [58]. Optical pistons, which consist of the piston rod, base piston, and piston crown, are occupied with a flat window entry as clarified in Fig. 12. These pistons are characterized by the following [59]:

- Piston crown is covered with a gasket ring that has a contact only to the metal liner part.
- Oil drainage on the base piston.
- Piston rods which have high mechanical stiffness.
- Lateral piston motion is minimized by using piston guide rings.
- Thermal piston expansion is accounted for by cylinder to liner clearance.

The piston crown is designed to vary according to GDI applications [58]. In addition, the gasket rings are installed on the piston crown's lower end and glide on the liner part of the metal only. Subsequently, this leads to a remarkable reduction in the engine compression ratio [60]. Figure 12 shows a schematic for the piston components in a GDI single-cylinder optical engine.

Piston rings are located on the lower position rather than on the normal position in OSCE. The use of this low ring position along with 0.5-mm clearance between the cylinder and linear reduces the compression ratio of about two units and adds to the combustion chamber's dead volume [61]. In diesel OSCE, this reduction in the compression ratio has a noticeable impact on ignition delay and combustion. However, in spark ignition (SI) engine, this reduction in the compression ratio is more acceptable as the main purpose of OSCE is to analyze the mixture formation during the compression and intake stroke [59].

The operation of an OSCE is performed using a modular testbed, which consists of different modules, as shown in Fig. 13. The major parts of such system are the dyno and engine; however, productive test operation usually consists of many different peripheral components such as [59] the following:

- engine and measurement modules with control of dyno;
- engine conditioning;
- control software and safety actuators;
- media conditioning and supply;
- fuel injection system; and
- user interface to access all actuators and signals.

Flame propagation in GDI configurations can be evaluated using OSCE that can provide the most useful information about flame propagation, diffusion, and combustion [59]. In OSCE, the features of the flame can be recorded with single-shot or high-speed cameras, which results in several flame imaging information as illustrated in Fig. 14. Diffusion and premixed flames can be easily distinguished because of their specific radiation features. Obviously, for engineering purposes, it is necessary to use an image that provides statistical information to make a decision about the best operating variables.

Thus, it is clear how OSCE can highly enhance many features in the field of engine combustion and emission, in addition to the contribution in different research and development applications. OSCE measurement instrumentation may consist of complex thermodynamics, optical, or emission diagnostic techniques. The efficient

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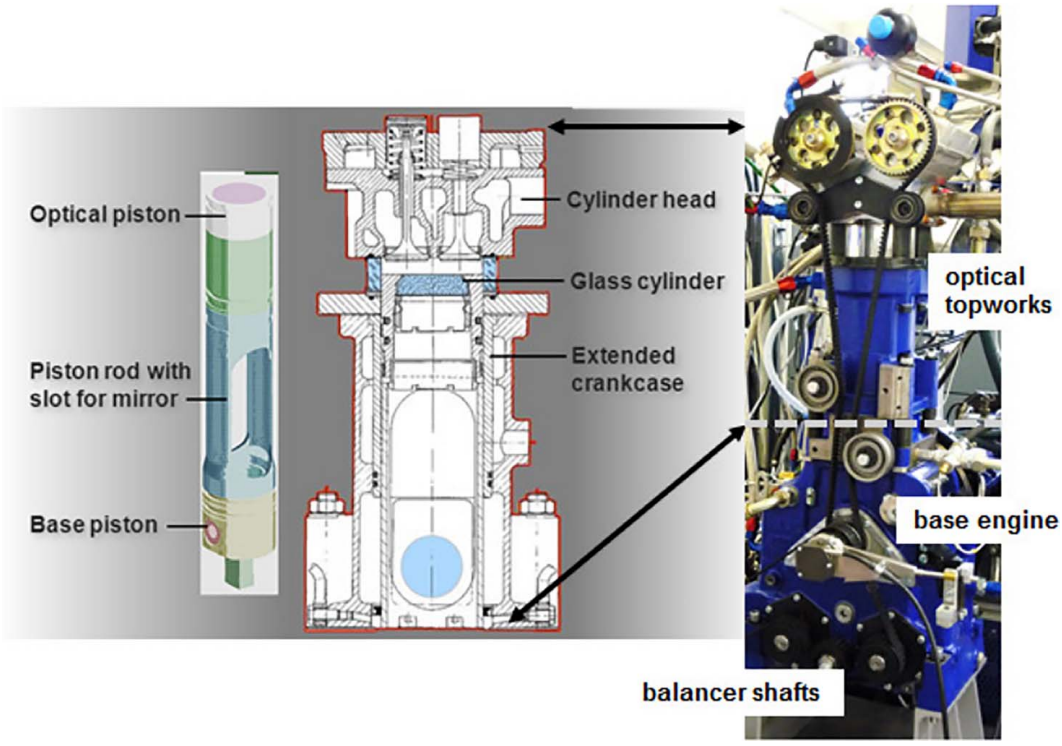

Fig. 11 Specific modules and complete assembly of an OSCE [58]

use of these complex systems can be simplified by the proper usage of the diagnostic methods and with the adaption of different modular arrangements in that platform.

3.1 Diesel Optical Engine. Diesel engines that are widely used as power generators, mechanical engines, or in mobile drives find an extensive spread use in construction equipment, automobiles, locomotives, and countless industrial applications. Their realm involves almost all industries and can be clearly observed daily at several applications [62]. Due to this importance, many researches are continuously developing in order to study their performance characteristics and lower their emission rates. One of the most reliable research tools to match this demand is the use of an optical single cylinder engine (OSCE), which has different visualization and measurement capabilities. Subsequently, it is considered as a powerful tool to predict, to a very close extent, the performance and emission characteristics of the real multi-cylinder diesel engine. Different researches have adopted this idea in several studies, and they have considered the use of OSCE to study the effect of varying some parameters on the engine's combustion characteristics. This subsection aims to lay a literature review on the most recent researches that have utilized diesel optical engine to conduct experimental studies in different aspects (ignition delay time, in-cylinder pressure, heat release rate, and flame luminosity).

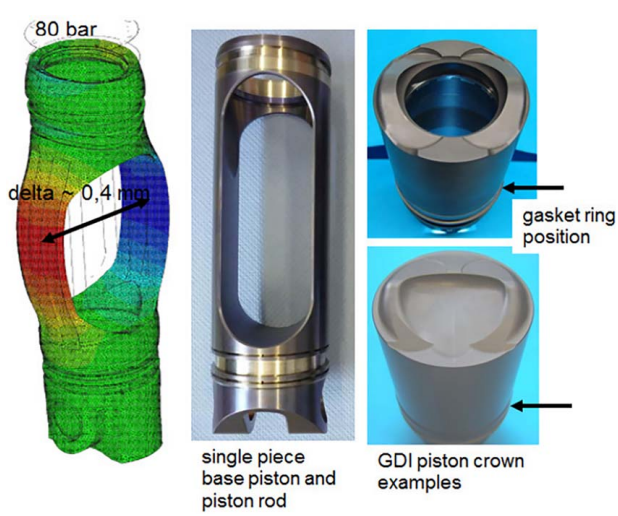

Fig. 12 Elongated piston for OSCE. Load analysis for the piston rod, piston components for a GDI application. Notice the low position of piston gasket ring [59].

3.1.1 Ignition Delay Time. Ahmad et al. [63] investigated the combustion characteristics of diesel–methane dual fuel (DF), more specifically the ignition delay time (IDT) and the combustion duration (CD) as a function of methane equivalence ratio (Φ_{CH_4}). Figure 15 shows the effect of varying methane equivalence ratio Φ_{CH4} on the ignition delay time at three different top dead center temperatures (T_{TDC}). It can be noticed that IDT increases as ϕ_{CH_4} increases due to the increasing effect of the inhibiting methane on the ignition of pilot diesel [64–66].

In the same principle, it was found that the CD increases when ϕ_{CH_4} in the mixture increases. However, the CD starts to decrease at $\phi_{CH_4} = 0.85$, which can be attributed to the existence of local fuel-rich zones at advanced (second and third) combustion stages. The combustion chamber is widened due to the presence of pilot diesel-rich zones, which causes the premixed methane–air mixture to start burning rapidly at the end of the second stage and during the high-intensity third stage.

Cheng et al. [67] evaluated the cycle-to-cycle variation (CCV) using an optically accessible heavy-duty diesel engine fueled with methane (main fuel) and diesel (pilot fuel). Figure 16 shows the variation of global intensity (bottom) and intensity-based coefficient of variance (COV) (top) at different λ_{CH_4}. It can be noticed that decreasing λ_{CH_4} in the fuel mixture results in a monotonic decrease in the global intensity. The addition of CH_4 has an increasingly high effect on high- and low-temperature reactions in CH_4–diesel–air

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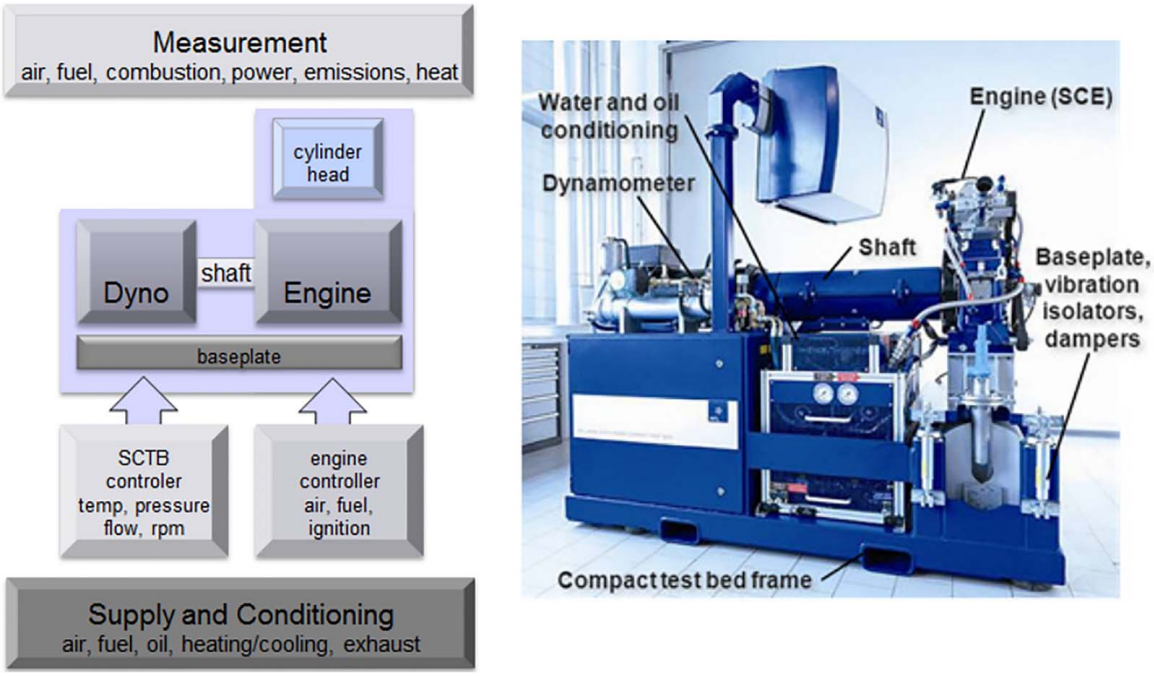

Fig. 13 Schematic of a testbed platform system, an example of a "compact testbed" configuration dedicated to the SCE operation [59]

mixture, which can result in more luminous soot formation and a longer IDT.

3.1.2 In-cylinder Pressure. Lee et al. [68] searched out the effect of varying the injection timing on the in-cylinder pressure rise using a diesel optical single-cylinder engine. In addition, a comparison between the experimental and simulation results was conducted. It was found that the maximum in-cylinder pressure decreases more when the injection timing is retarded and the simulation results have approved well the same. Cheng et al. [67] evaluated the cylinder-pressure as a function of λ_{CH_4} using an optically accessible heavy-duty diesel engine fueled with methane (main fuel) and diesel (pilot fuel). It was observed that CH_4 plays as an inhibiting factor for combustion in the mixture. As a result, adding more CH_4 into the mixture causes the peak pressure to be higher. Pastor et al. [69] measured the cylinder's mechanical deformation of a diesel optical single-cylinder engine at two different intake pressures (1.1 and 2.7 bar) that correspond to two different top dead center (TDC) cylinder pressures (31 and 80 bar), respectively. When the intake pressure was 1.1 bar, the maximum in-cylinder pressure was found to be 31 bar with a 0.03 mm deformation at TDC. It was noticed for this condition that the mechanical deformation and the in-cylinder volume were almost negligible. On the other hand, when the intake pressure was selected to be 2.7 bar, the maximum in-cylinder pressure was 80 bar with a maximum TDC deformation that is equal to 0.5 mm. It can be deduced that as the in-cylinder pressure rises, the mechanical deformation becomes more significant. For the latter configuration, the deformation coefficient was found to be 2.25.

3.1.3 Heat Release Rate. Ahmad et al. [63] stated that dual-fuel combustion is a complex process to analyze, and it depends on the chemical and physical properties of pilot-fuel sprays, in addition to the concentration of each gas in the premixed mixture. In diesel–methane DF combustion, pilot diesel is injected during the compression stroke close to the TDC, which provides sufficient

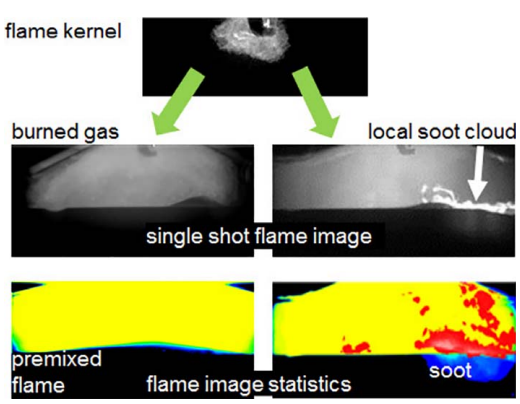

Fig. 14 Flame analysis shows a premixed combustion situation (left hand) and appearance of soot clouds because of wall film combustion. Imaging and data evaluation procedures provide a statistic image evaluation [59].

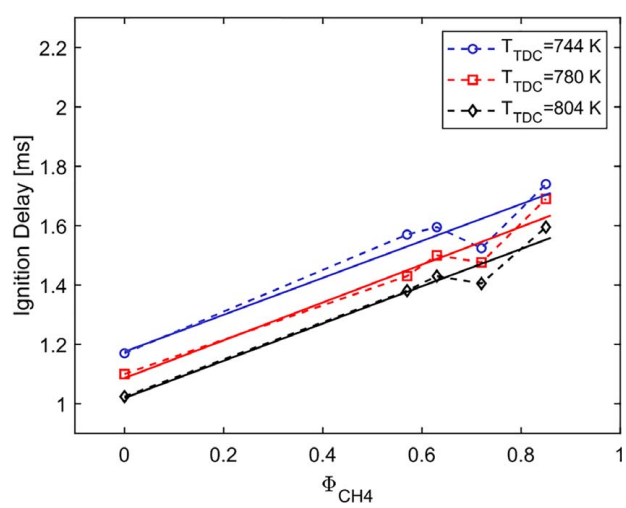

Fig. 15 IDT trend at three different TDC temperatures [63]

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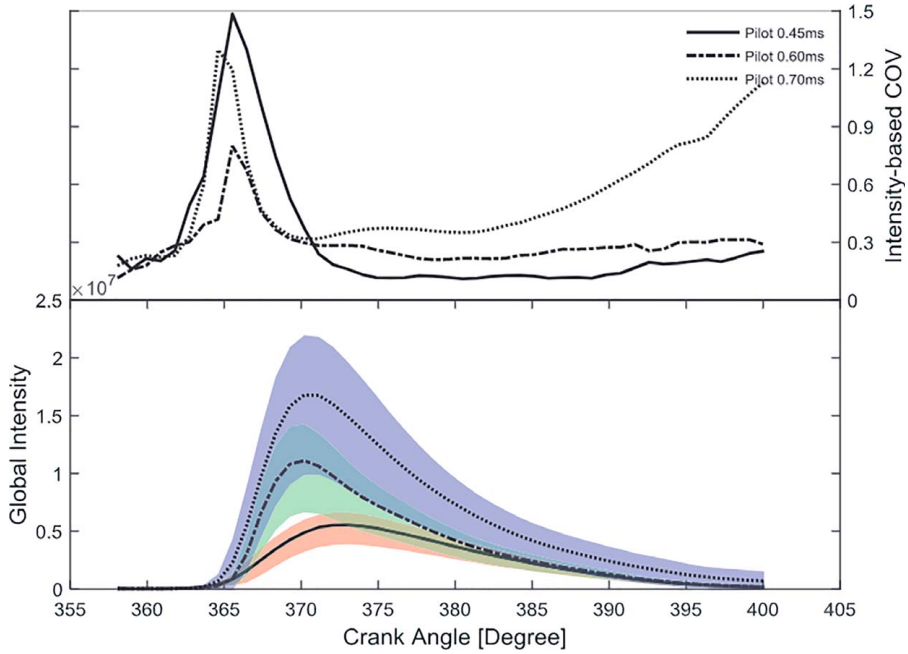

Fig. 16 Effect of pilot fuel ratio on the global intensity and intensity-based COV [67]

energy for the ignition of the premixed mixture. Subsequently, the characteristics of heat release constitute complex chemical and physical interactions. Therefore, the process of DF combustion can be viewed as a combination of three overlapping stages as clarified in Karim's model earlier Karim et al. [70]:

(1) First-stage combustion, θ_1, is due to the combustion of some premixed methane–air mixture in addition to pilot diesel.
(2) Second-stage combustion, θ_2, is due to the combustion of premixed methane–air mixture in the surrounding of pilot diesel.

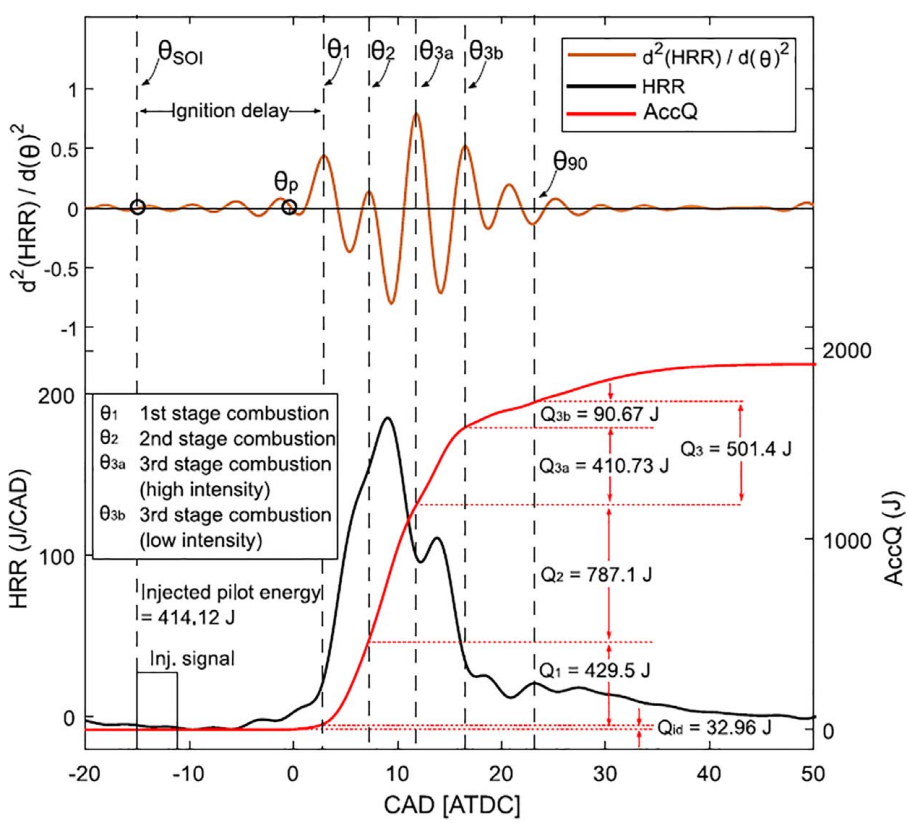

Fig. 17 Characterization and identification of three overlapping combustion stages of a typical DF combustion representative case by the second derivative of HRR: CAD, crack angle degree

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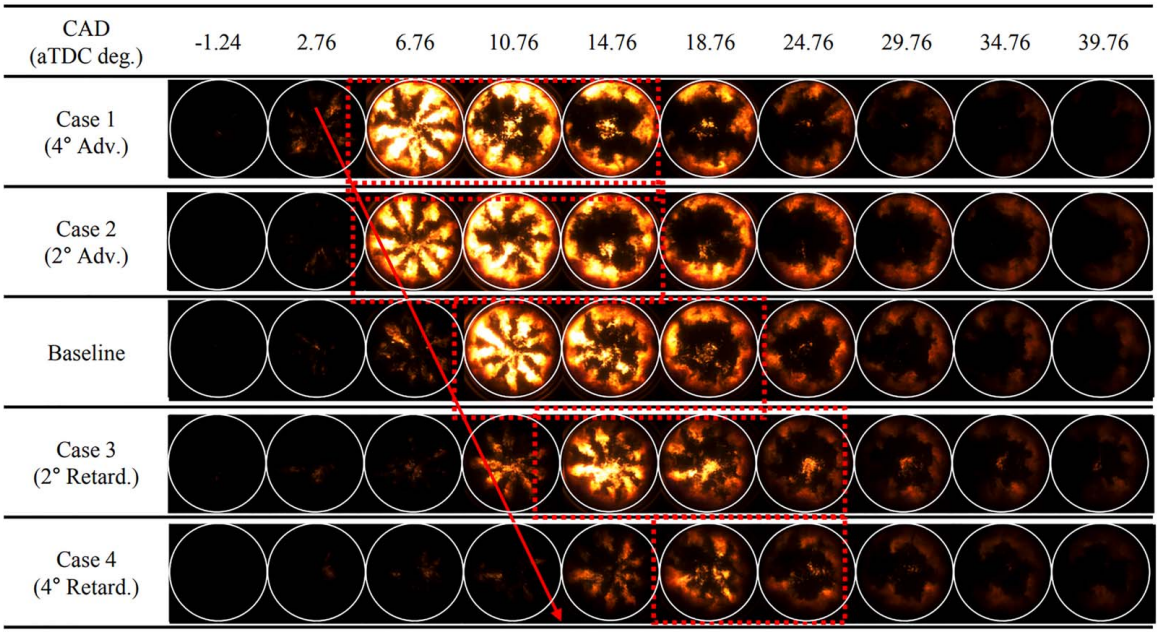

Fig. 18 Flame luminosity as a function of injection timing sweep for five different cases [68]

(3) Third-stage combustion, θ_3, is caused by the turbulent flame propagation with the remaining diluted unburned methane–air mixture.

Some studies have identified the DF combustion process depending on combustion modes and stages, requesting further investigation [71–74]. The three overlapping stages could be identified and separated using the second derivative of the heat release rate (HRR) in which features of natural luminosity (NL) images recognize combustion mechanisms. The second derivative of HRR indicates for the inflection points, which is more challenging to highlight using only the first derivative and desires further thresholding. A typical DF combustion HRR profile is illustrated in Fig. 17 along with the accumulate heat release (AccQ) in addition to the second derivative of HRR (SDHRR). The local maximum value of SDHRR is θ_1, θ_2, θ_{3a}, and θ_{3b}, indicating for first-, second-, and third-stage combustion, respectively.

Ahmad et al. [63] investigated the effect of ϕ_{CH_4} and T_{TDC} on DF combustion and obtained HRR and AccQ using the same combustion model. HRR profiles were observed to have multiple peaks (M-shaped), which vary in their magnitude depending on the amount of CH_4 in the fuel mixture. Such profiles of HRR have been reported in different investigations [71–75]. As ϕ_{CH_4} is increasing, the shape of the HRR curve is brought to change from multiple peaks (M-shape) gradually to a quasi-single peak (bell-shaped). Meanwhile, the combustion stages remain to interact with the increase of ϕ_{CH_4}. The HRR data were analyzed using the second derivative method to make the study deeper and expand the influence of increasing ϕ_{CH_4} on the characteristics of DF heat release. The combustion stages θ_1, θ_2, θ_{3a}, and θ_{3b} along with θ_{90} were highlighted on the respective AccQ curves for every ϕ_{CH4} case at several T_{TDC} [76].

Cheng et al. [67] evaluated the HRR as a function of λ_{CH_4} using an optically accessible heavy-duty diesel engine fuel with methane (main fuel) and diesel (pilot fuel). The inhibiting effect for the addition of CH_4 was clearly observed in HRR. This can be referred to the weak flammability of the air–CH_4 mixture for the ultra-lean CH_4 condition. There is unstable combustion and a slow flame speed, which results in a lower HRR.

3.1.4 Flame Natural Luminosity. Lee et al. [68] used the visualization kit that lies between the cylinder head and the block to perform visualization experiments. It was found that the flame luminosity increases more when the injection timing is advanced. Figure 18 illustrates the variation of NL as a function of the crank angle at five different injection timings (including the baseline case).

Ahmad et al. [63] gathered intensity distribution and NL images that take place during the second-stage combustion θ_2 (Fig. 19). A high-speed camera (Photron Fastcam APX-RS 250 K) that is

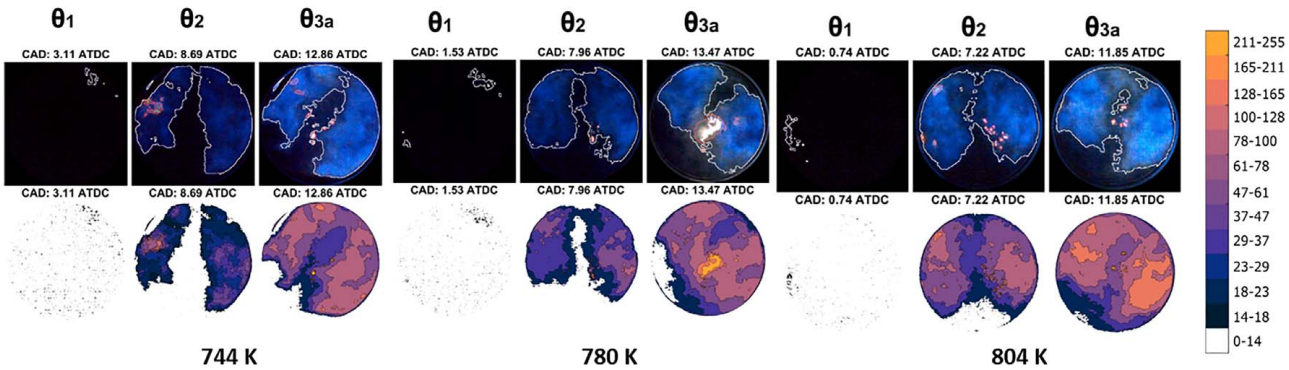

Fig. 19 Representative NL images and intensity distribution images immediately after the start of the combustion stages for the entire range of test points investigated in the study (effect of ϕ_{CH_4} and T_{TDC} on DF combustion)

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equipped with a Nikon AF-Nikkor 70 mm/3.5 objective lens recorded temporally and spatially resolved NL images. As the increased temperature plays a vital role in enhancing the pilot-diesel evaporation rate [77,78], there is almost no, or few local fuel-rich zones visualized near the periphery when $\phi_{CH_4} = 0.57$ and 0.63 at higher temperatures (O_2 concentration is slightly high). However, it was observed that when $\phi_{CH_4} = 0.72$ and 0.85, even at higher temperatures, local fuel-rich zones are visualized to be intense owing to inadequate O_2 concentration, which leads to more soot formation. Moreover, fuel-rich zones in the middle of the combustion chamber are assumed to result from injector dribbling [63]. For more details about flame propagation and dribbling in DF combustion, the reader is directed to other previous researches [63,78,79].

Summary

A summary of studies using diesel optical engine is provided in Table 2 for the convenience of the reader.

3.2 Gasoline Optical Engine. Gasoline optical engines behave as a representative tool in the research and development of new concepts for combustion in SI engines. This is due to their capabilities in allowing for the application of quantitative and qualitative diagnostic techniques, which can be used to study combustion and emissions, mixing, and in-cylinder flow characteristics [80]. Such experimental measurements are not only essential for validating the computational model; however, they can also be processed to provide detailed information about the physical processes that take place inside the cylinders. Subsequently, they are useful in developing new combustion strategies as the GDI [81]. In this context, the role of this section comes to give a glance about the past researches, which have utilized gasoline optical single-cylinder engine as a research tool to perform combustion studies.

3.2.1 In-cylinder Pressure. Boggio et al. [82] used an accessible optical single-cylinder spark-ignition engine to carry out measurements that show the effect of adding low levels of hydrogen to methane (5%, 10%, and 20% volumetric basis) on the in-cylinder pressure under lean-burn and stoichiometric conditions. Thermodynamic results illustrated an increase in the maximum in-cylinder pressure as more hydrogen is added to methane fuel, and this effect becomes more evident while operating the engine in the leanest condition. Giovani et al. [83] have also studied the combustion characteristics using thermodynamic (heat release) depicted from in-cylinder pressure measurements. Three different fuels were used in the optically accessible single-cylinder gasoline engine: pure gasoline (G100), pure ethanol (E100), and 50% ethanol/gasoline blend (B50). As expected, the use of pure ethanol has resulted in the highest maximum in-cylinder pressure at both operating conditions, while the lowest belongs to pure gasoline. In addition, it was observed that the time at which the peak in-cylinder pressure is reached is in advance for pure ethanol when compared with other tested fuels. Consequently, it can be deduced that faster combustion than other fuels characterizes ethanol. Considering the 50% blend, it behaves as noticed with intermediate characteristics in comparison with pure fuels; however, during the combustion early stages, it shows trends that are closer to the pure ethanol fuel.

Id et al. [84] investigated the combustion characteristics of gasoline fuel at five different air-fuel ratios utilizing an optically accessible direct injection spark ignition (DISI) single-cylinder engine. Lean and stoichiometric operations were studied through a combined optical and thermodynamic approach. The results of the average in-cylinder pressure at five different fuel–air ratios were obtained using a pressure sensor. The highest peak pressure was detected during the stoichiometric operation and during the expansion process. The operation of the engine at points close to maximum brake torque (MBT) can be evidenced by the location of the peak pressure that is reached with the same range for all air-fuel ratios.

3.2.2 Flame Propagation Rate. Yang et al. [85] designed two tumble deflectors that have two different tumble ratio (TR = 1.5 and 2.2) using computational fluid dynamics and computer-aided design. Afterward, they were 3D-printed and then installed on the optical engine intake ports, and then, the firing tests were performed (Fig. 20). The tumble deflectors were used to direct the flow into the intake ports so that changing the deflector thickness causes a variation in the level of the tumble.

The use of the first (TR = 1.5) and second (TR = 2.2) tumble deflectors have resulted in a 13.9% and 34.5% increment in the flame propagation rate, respectively (Fig. 21). This can be interpreted by the increase in TKE, which facilitates the mass-heat transfer between the unburnt and burnt zone, which results in a shorter combustion duration and an earlier heat release. Besides, a

Table 2 Summary of literature survey on the diesel optical engine

	Diesel optical engine			
Reference	Ahmad et al., 2019 [63]	Cheng et al., 2019 [67]	Lee et al., 2019 [68]	Pastor et al. 2018 [69]
Fuel	Diesel/methane	Diesel/methane	Diesel/air/EGR air	Diesel
Equivalence ratio	0.57, 0.63, 0.72, 0.85	0.50, 0.56, 0.625	–	1.0
Engine temperature (K)	353	298, 318, 328	–	323
Engine speed (rpm)	1400	1400	1500	1200
Compression ratio	13.5:1	17.9:1	16.0:1	13.7:1
Intake air pressure (bar)	1	1.1	–	1.1, 1.5, 1.9, 2.2, 2.7
IMEP (bar)	9–10	10	6	–
Swirl ratio	2.7	2.7	2.4	–
Optical access	Bowditch-designed optical access	Bowditch-designed optical access	Bowditch-designed optical access	Bowditch-designed optical access
Pressure sensor type	Kistler type 6125C	Kistler type 6125C	Kistler type 6125C	Kistler 4603B10
Camera type	Photron Fastcam APX-RS 250 K	Photron Fastcam APX-RS)	Mini AX-100, PHOTRON	Photron SA-X2
Frame rate	9000	9000	100	21,000
Main measurements	Heat release rate Natural luminosity Ignition delay time Combustion duration	Pressure-based COV Heat release rate In-cylinder pressure Global intensity	In-cylinder pressure Heat release rate Ignition delay time Combustion duration	Piston displacement Heat release rate In-cylinder pressure In-cylinder volume
CAD at injection	356 deg aTDC	−15 deg aTDC	−22.24 deg aTDC	–
CAD at exhaust	150 deg aTDC	–	126 deg aTDC	–
No. of averaged cycles	150	150	–	50

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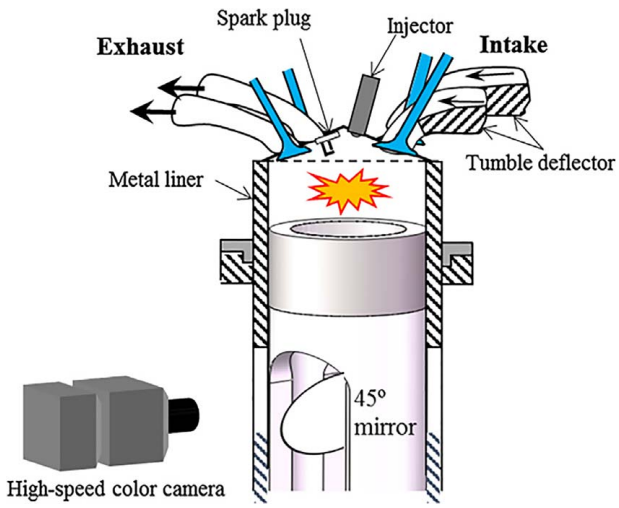

Fig. 20 Experimental setup for the high-speed imaging of in-cylinder combustion [85]

delayed and less appearance of yellow flame was originated, which indicates a reduced soot formation.

Boggio et al. [82] also investigated the flame growth speed and flame development for both ethanol and gasoline. It can be clearly observed from Fig. 22 that ethanol fuel is characterized by faster combustion when compared to gasoline. Moreover, the flame growth speed was obtained for both fuels by deriving the interpolating curves, which further confirmed the quick development of ethanol flame upon ignition.

3.2.2.1 Indicated Mean Effective Pressure. Chen et al. [86] investigated the effect of using different spark plug types and spark ignition energy on the optical engine indicated mean effective pressure (IMEP) that operates with natural gas. Synchronous flame images and in-cylinder pressure track were used to discuss early flame development and cycle-to-cycle variations while addressing the role of ignition energy and spark plugs. As for IMEP, it is evident in Fig. 23 that there is a particular crank angle where the highest IMEP is achieved for each spark plug configuration. The maximum IMEP is 5.87 bar obtained for the single-claw spark plug at a spark timing of −20 °CA, while the maximum IMEP (IMEP = 6.32) for the multi-claw spark plug is achieved at a spark timing of −14 °CA. Thus, it is indicated that when running the engine under 50 mJ ignition energy, the thermal efficiency of the engine can be improved with the use of a multi-claw spark plug as it can enhance the combustion stability and hence optimizes the combustion phase.

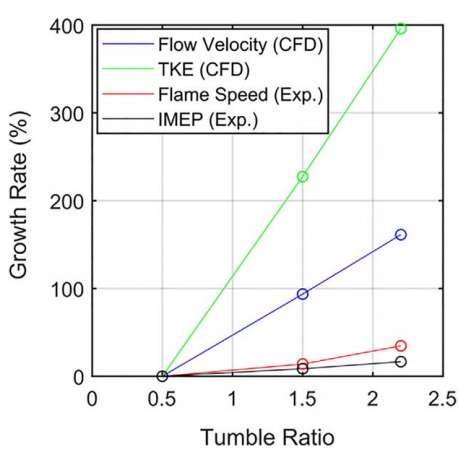

Fig. 21 Growth rate of flow velocity and TKE (spatial averaged at the spark timing of 15 deg bTDC), as well as flame speed (characterized as flame area at 7.15 deg aTDC) and indicated mean effective pressure (IMEP) with increased tumble ratios. Flow velocity and TKE are acquired from the computational fluid dynamics model, while flame speed and IMEP are experimental results [85].

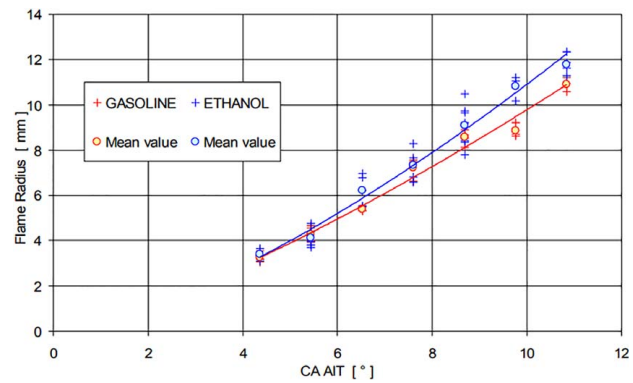

Fig. 22 Flame radius for gasoline and ethanol at different crank angles [82]

Yang et al. [85] also considered the effect of using the three different tumble deflectors on the optical engine IMEP from 100 consecutive cycles. It was noticed that the average IMEP of TR = 0.5 was about 2.59 bar, and with the use of the tumble deflectors, it increased to 2.81 bar and 3.02 for the TR = 1.5 and TR = 2.2 deflectors respectively. This can be referred to the increased TKE and flow velocity found in the same work. Besides, this can also be explained by the faster flame propagation with the use of the tumble deflectors.

Id et al. [85] found that as the air-fuel ratio was augmented, the IMEP has decreased as the amount of fuel injected was reduced in comparison with the stoichiometric case. At the limit of lean combustion ($\lambda = 1.6$), COV was found to be reduced by around 30% concerning the stoichiometric condition ($\lambda = 1.0$).

3.2.2.2 Unburned Hydrocarbons and NO_x Emission. Boggio et al. [82] have considered the analysis of exhaust gases, namely, UHC (unburned hydrocarbon) and NO_x, as they provide valuable information about the chemistry of combustion. At the same air-to-fuel ratio, UHC emissions decrease when adding more hydrogen in the blend, which is the same conclusion found in another study [87]. The reason behind that can be referred to the reduction of methane volume percentage in the air–fuel mixture.

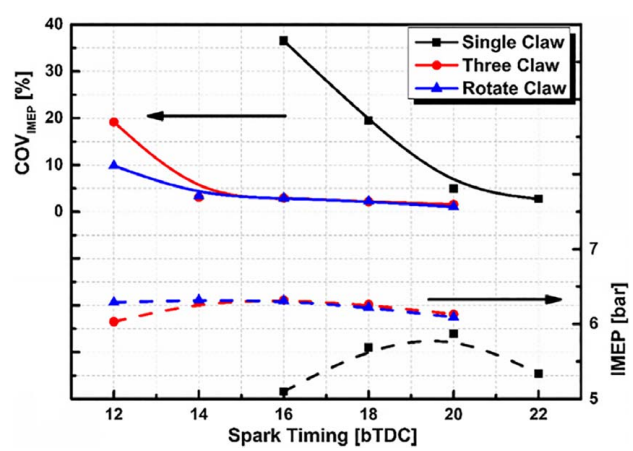

Fig. 23 Covariance of indicated mean effective pressure (COVIMEP) and indicated mean effective pressure (IMEP) observed with three spark plug types and 50 mJ ignition energy [86]

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Table 3 Summary of literature survey on the gasoline optical engine

	Gasoline optical engine			
Author	Peñaranda and Boggio, 2018 [82]	Yang et al., 2019 [85]	Chen et al., 2019 [86]	Id et al., 2017 [84]
Fuel	Methane/hydrogen blends	95 Gasoline	Methane	Gasoline
Equivalence ratio	1.0, 0.71	1.0, 0.71	1.0	1, 0.77, 0.71, 0.67, 0.625
Engine temperature (K)	330–335	298	368	330–335
Engine speed (rpm)	900	1200	800	1000
Compression ratio	–	11:1	13:1	10:1
Optical access	Bowditch-designed optical access	Bowditch-designed optical access	Bowditch-designed optical access	Bowditch-designed optical access
Pressure sensor type	Quartz pressure transducer	Kistler type 6125C	Kistler type 6125C	–
Camera type	PCO Dimax S1	High-speed color NAC CMOS	Photron SA-Z	Optronis CamRecord 5000
Frame rate	5400	–	5000	5000
Main measurements	In-cylinder pressure UHC, CO, and NO_x emission Flame propagation rate Maximum flame penetration	Tumble ratio Turbulent kinetic energy Swirl ratio Indicated mean effective pressure	In-cylinder pressure Mass fraction burned Flame development Indicated mean effective pressure	Fuel conversion efficiency In-cylinder pressure Heat release rate Indicated mean effective pressure
CAD at spark	−14 CAD ATDC	15 CAD BTDC	−18 CAD ATDC	10,15,20,25,30 CAD BTDC
No. of averaged cycles	200	100	150	200

Subsequently, the composition of the unburned gas that is trapped in the crevices reduces the methane emission rate. Besides, there is a shorter quenching distance for hydrogen when compared with methane. The increase in the air-to-fuel ratio results in reduced overall combustion efficiency, and hence, UHC emission was higher for the same proportion of hydrogen.

Increasing the amount of hydrogen in the blend causes an increase in NO_x emission rates, especially for the stoichiometric case. As expected, when operating at the leaner case, the NO_x concentration has significantly decreased. This can be directly related to the reduced in-cylinder temperature and pressure that were found in the same work. A significant result that can be also extracted from the same study that the lean combustion process can drastically decrease NO_x emission, and that was clearly noticed when running the engine at $\lambda = 1.0$ case.

Id et al. [84] have reached the same conclusion when operating the gasoline optical engine with different air–gasoline fuel ratios. As the lean flammability limit was approached, an increased amount of hydrocarbon (HC) emission has been detected. This can be related to the prominent phenomenon of quenching close to the cylinder walls. Nonetheless, this phenomenon is responsible for reducing the overall efficiency.

The highest value for NO_x was reordered when operating with the stoichiometric combustion condition. The high evolution of nitrogen oxides at that condition can be related to the thermal effect [88]. When operating with lean mixtures, two competing factors (low-burned gas temperature and high concentration of oxygen) which possess different weights, taking into consideration that temperature has a more dominating effect result in less NO_x formation. NO_x concentration was found to be reduced by 95% when operating at the leanest case, and this value is very close to the efficiency of a three-way catalytic converter [89]. Thus, it is highly emphasized to run the engine with a lean operation to reduce emissions and increase the efficiency, while paying attention to the use of catalytic converter in reducing HC and CO when operating at this condition [90].

Summary

A summary of studies using a gasoline optical engine is provided in Table 3 for the convenience of the reader.

4 Conclusions

It is clear from the above discussions that turbulent and laminar flames of different fuels have been studied extensively by using a fan-stirred cylindrical/spherical combustion bomb. This technique has proved to be a reliable one in measuring the laminar and turbulent premixed flame speeds at different initial conditions. The combustion bomb technique allows for easy-control of the initial temperature, pressure, and turbulent intensity. The latter can be controlled by changing the fan geometry and speed. In addition, previous investigations have shown that flame visualization can be done using Schlieren imaging and laser tomography. However, none of the previous studies has considered the use of OH-chemiluminescence as a measurement technique despite being one of the efficient available techniques. Consequently, this measurement tool can be used for tracking flame development under turbulent conditions in future research. Moreover, the combustion bomb technique is flexible in measuring flame speeds of different kinds of gaseous and liquid fuels such as alkanes (C_1-C_3), natural gas, biogas, and isooctane.

An optical engine is a new and advanced method to investigate turbulent flames in realistic practical conditions. Recently, diesel and gasoline optical engines have been used in few investigations to study the flame and combustion characteristics of different fuels. For example, diesel optical engines were used to study ignition delay time, in-cylinder pressure, and heat release rate for a number of heavy diesel fuels. In addition, flame visualization and luminosity inside the engine have been observed to study the effect of the injection timing on flame propagation and characteristics. Similarly, gasoline optical engines have been considered as an excellent tool to investigate deeply the turbulent premixed flame behavior in actual conditions. This is due to their capabilities in allowing for the application of quantitative and qualitative diagnostic techniques, which can be used to study combustion, emissions, mixing, and in-cylinder flame characteristics. Such experimental measurements are not only essential for validating the computational model but can also be processed to provide detailed information about the physical processes that take place inside the cylinders.

As Qatar is the largest producer and exporter of GTL, it is highly important to measure its combustion characteristics, which have not been fully studied until now. Subsequently, this demand can be met by using the cylindrical/spherical combustion bomb technique, and the results can be deepened and validated using the optical engine as a strong flame visualization tool.

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